

SAIF UR REHMAN

Simulation Engineer
Automation & Scientific Development

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[Personal Portfolio](#)

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ABOUT ME

Simulation engineer with 3+ years across industrial CAE and research, specializing in high-fidelity CFD and structural multiphysics for aerodynamic bodies (airfoils, drones, external flows) and in automating the full simulation workflow through scientific code. Core developer of a hybrid C++/Python adjoint-based shape-optimization framework; built Python pipelines that run 200+ HPC simulation campaigns end to end.

TECHNICAL SKILLS

- **Programming & Scientific Computing:** Python (NumPy, pandas, SciPy, scikit-learn, gmsh), C++ (OOP, performance kernels), Bash; GNU parallel, HPC clusters, Linux/OpenMPI, CI & testing pipelines, data pipelines
 - **Optimization & MDO:** Adjoint methods, gradient-based optimization, vertex morphing, multi-objective / multi-constraint design, solver coupling, KratosMultiphysics
 - **CFD:** ANSYS Fluent, OpenFOAM, external aerodynamics, steady/URANS, RANS–VOF multiphase
 - **FEA & Structures:** Abaqus, LS-DYNA, MSC Nastran, OptiStruct, ANSA, META;
 - **AI / Surrogate Modelling:** Kolmogorov-Arnold Networks (KAN), Gaussian Process/Kriging, Random Forest, MLP/ANN, SVM, DOE (Latin Hypercube), Sobol sensitivity
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WORK EXPERIENCE

Simulation Engineer

(Dec 2023 - present)

SOCO Engineers, Multan, Pakistan

- Built Python automation pipelines covering the pre/postprocessing workflows - geometry/mesh preprocessing, HPC-enabled solving, and post-processing - cutting setup time by ~30% across 20+ NVH and crash load cases
- Core-developed a node-based adjoint-driven shape-optimization framework for multi-objective, multi-constraint design, architecting a hybrid system of C++ compute kernels (mapping, search, linear algebra) and a Python orchestration layer for solver coupling and workflow management; integrated Abaqus, OptiStruct, STAR-CCM+ and KratosMultiphysics.
- Ran large-scale simulation campaigns on HPC clusters (Linux/OpenMPI), extracting and managing structured datasets from 200+ runs to train surrogate models.
- Executed CFD and multiphysics studies end to end and extended them to coupled fluid–structure interaction (FSI), capturing structural response of components under external-aerodynamic and flow loading.
- Engineered a materials-modelling workflow over a central database of 300+ supplier materials, running virtual material testing (uniaxial, multiaxial, torsion, buckling) via explicit dynamics for flow-curve and yield-locus calibration.
- Resolved 30+ reported issues in simulation model setup and HPC configuration, and built a "simulation bench" that version-controls multiple variants, input decks and subvariants across design phases.

Research Assistant

(Nov 2022 - Dec 2023)

National University of Sciences and Technology (NUST), Islamabad, Pakistan

- Developed and validated high-fidelity CFD models for turbulent, multiphase, cavitating and external-aerodynamic flows, resolving separation/reattachment, wake dynamics and transport phenomena to within <8% of experimental and published benchmarks.
- Ran parametric and mesh-independence studies across Reynolds numbers up to 18,000, quantifying geometry and operating-parameter effects on aerodynamic and thermal-hydraulic performance.
- Integrated a Kolmogorov-Arnold Network (KAN) surrogate with unsteady CFD time-series data for real-time drag prediction, reaching $R^2 \approx 0.99$ with millisecond inference and outperforming ANN and SVM baselines.
- Authored/co-authored 3 peer-reviewed publications and technical reports on CFD-based fluid and multiphysics systems (see Publications).

PROJECTS

NACASurrogate - Automated Aerodynamic Surrogate for NACA Airfoils

OpenFOAM 12, gmsh Python API, Jinja2, pyDOE2, scikit-learn, SMT, GNU parallel, HPC, SALib

- Built a fully automated CFD-to-surrogate pipeline (DOE → geometry → meshing → solve → harvest → train → validate) driving 200+ RANS runs across four validated flow regimes, mapping (α , Re, thickness) → (Cl, Cd).
- Trained and benchmarked 4 surrogate models (Gaussian Process, Random Forest, MLP, Kriging) to $|Cl| \leq 5\%$, $|Cd| \leq 10\%$, with automated Sobol sensitivity and mesh-quality gates (checkMesh, $y^+ = 0.5$).

Adjoint-Based Shape-Optimization Framework

C++, Python, vertex morphing, adjoint solvers, CFD/FEA coupling

- Contributed in industrial adjoint optimization framework spanning fluid, structural and multiphysics problems; C++ kernels for compute-intensive tasks and Python for orchestration across 4+ coupled solvers, using vertex morphing for smooth, mesh-consistent geometry updates.

Ventilated Supercavitation Drag Reduction with KAN Surrogate

ANSYS Fluent (URANS, VOF multiphase), Python, KAN/ANN/SVM

- Validated unsteady CFD of cavity and wake dynamics achieving up to 26% drag reduction (~16% lower propulsion power); coupled a KAN surrogate at $R^2 \approx 0.99$ for real-time drag evaluation.

Unsteady Vortex-Shedding around Underwater Cylinders

ANSYS Fluent (URANS-SST), Python, ICEM-CFD

- Resolved unsteady wake and vortex-induced vibration (VIV) at $Re = 10,000$, with spectral (frequency) analysis validated against experimental benchmarks - directly relevant to structural fatigue and flow-induced-vibration assessment.

PENDULA - N-Link Chaotic Pendulum Solver (scientific code)

Python, Lagrangian mechanics, SciPy solve_ivp, Matplotlib

- Wrote a generalised N-body dynamics solver deriving equations of motion symbolically for arbitrary N and integrating with energy-conservation checks — demonstrating from-scratch numerical-methods and scientific-code development.

EDUCATION

Bachelors in Mechanical Engineering

School of Mechanical and Manufacturing Engineering (SMME),
NUST, Islamabad

(2019 - 2023)

Relevant coursework: Computational Fluid Mechanics, Numerical Methods, Finite Element Analysis, Heat Transfer

Final-Year Thesis: Design & Development of an Agricultural Drone with ANN-Based Path Prediction
(UAV design, prototyping, system integration)

CERTIFICATIONS

3-Week OpenFoam Workshop -- OpenFOAM Organization

(Oct 2022)

Mastering ANSYS, Level 1 & 2 -- Udemy

(Nov 2021)

Training Certificate on CFD OpenFoam -- SMME NUST

(Oct 2022)

Neural Networks and Deep Learning -- Deeplearning.ai

(Dec 2024)

PUBLICATIONS / TECHNICAL REPORTS

[1] U. Farooq, **S. Rehman**, H. H. Riaz, et al., "Synergistic effects of Al₂O₃–ethylene glycol/water nanofluids and artificial roughness on solar heater thermal performance," Journal of Thermal Analysis and Calorimetry, vol. 150, pp. 16501–16517, 2025. doi: 10.1007/s10973-025-14571-4

[2] U. Farooq, **S. Rehman**, M. Taseer, et al., "Kolmogorov-Arnold Network Enhanced CFD Modeling of Drag Reduction in Ventilated Super-Cavities Around Underwater Projectiles," Engineering Applications of Artificial Intelligence, 2025 (under review).

[3] **S. rehman**, M. Malik, M. Nafay "Design and Development of Variable rate Sprayer for Agricultural Drones, Intelligent path predictor based on Artifical Neural Network", Bachelors Final Year Project.